Discussion on Data Quality Dimensions based on Article Designing Data Governance

Data governance is acknowledged as one of the successful practices to realize the value of data as an asset. Furthermore, in order to produce a single version of truth and comply to legislative mandates, data governance program need to be put into action. Thus, it has become a hot topic among data practitioners to discuss in global conferences (Khatri & Brown, 2010)

Based on the article entitled Designing Data Governance, Khatri and Brown highlighted data quality as one of the five data decision domains in their data governance framework. In the context of data quality, an organization should define the requirements for the data to fit-for-purpose. It means that they should specify the data quality dimensions that are relevance with the intended use of data. Besides, the locus of accountability should also determine the acceptable standards for data quality based on the dimensions, ways to communicate data quality and the evaluation procedures for data quality (Khatri & Brown, 2010).

Four dimensions of data quality that have been emphasized in the article are:

Dimension	Description	Example
Accuracy	The recorded data is reflecting the facts.	A patient with blood type A should be recorded as A in medical record, not O or other blood type
Timeliness	The recorded data is kept up to date.	A trading system should update the stock prices in real-time without delay.
Completeness	The recorded data cannot have missing values if it is a mandatory data.	Column name cannot be blank if the value is mandatory.
Credibility	The recorded data should come from trusted sources.	Data from an established articles is more trusted than unverified blog.

Uncovering 15 Additional Data Quality Dimensions through Literature Reviews

There is an indefinite numbers of data quality dimensions mentioned in books, reports and articles make it hard for the data practitioners to determine the best data quality dimensions to measure if their data fit for purpose. Thus, (Haug, 2021) conducted an extensive literature review that consolidates all data quality dimensions mentioned in the journal and conference articles found in Web of Science and Scopus databases. He identified more than 300 distinct data quality dimensions from more than 500 articles.

However, some terms may be overlapped as it has similar meanings such as "interpretability" and "comprehensibility". In addition, some similar terms may have different definitions. For example, some may define "accuracy" as percentage of data without errors while some may define as the degree of data representing real-world value (Haug, 2021).

In this section, a selection of 15 data quality dimensions extracted from seven articles are discussed as below. Some data quality dimensions presented may have more than one names but they represent the similar dimensions as their definitions means the same thing.

1. **Accessibility / Utility**

   Accessibility or Utility dimension means that the data is easy to access and retrieve or helpful for people with some disability. For example, the data has special configurations for people with vision impairment such as enlarged print.

Data Quality Dimensions	Definitions
Accessibility	i. "Extent to which information is available, or easily and quickly retrievable" (Sidi et al., 2012). ii. "The degree to which data can be accessed in a specific context of use, particularly by people who need supporting technology or special configuration because of some disability" (Miller et al., 2024). iii. "The dimension depicts the extent to which the data are available and with which ease the user can access them" (Hassenstein & Vanella, 2022).

	iv. "Accessibility measures how easy it is to acquire data when needed, how long it is retained, and how access is controlled" (Myers & P. Blake, 2017).
Utility	i. "The degree to which data can be accessed in a specific context of use" (Liu et al., 2020).

2. Consistency / Concordance

Consistency or Concordance means that the data is not contradicted and in the same format with previous data. For example, the date must be registered in the same format. 6/5/2024 is not consistent with 6 May 2024, thus it will flag as inconsistent data.

Data Quality Dimensions	**Definitions**
Consistency	i. "The extent to which data is presented in the same format and compatible with previous data " (Sidi et al., 2012). ii. "The degree to which data has attributes that are free from contradiction and are coherent with other data in a specific context of use. It can be either or both among data regarding one entity and across similar data for comparable entities" (Miller et al., 2024). iii. "The degree to which data are free of contradiction with themselves or follow an established rule, and are provided in the same format compatible with previous data" (Hassenstein & Vanella, 2022). iv. "Consistency measures whether or not data is equivalent across systems or location of storage" (Myers & P. Blake, 2017).

Concordance	i. "The degree to which data has attributes that are free from contradiction and are coherent with other data in a specific context of use." (Liu et al., 2020).

3. **Relevancy**

Relevancy means that the data fulfil the requirements or meet the expectation of the user and helpful for the task at hand. For example, to plan for marketing campaign, data that are related to customer demographics data such as gender, age, and occupation are relevant. Meanwhile, data about weather is not relevant.

Data Quality Dimensions	**Definitions**
Relevancy	i. "Extent to which information is applicable and helpful for the task at hand" (Sidi et al., 2012). ii. "The degree to which the data meets the expectations and requirements of the user" (Hassenstein & Vanella, 2022).

4. **Validity / Compliance**

Validity or Compliance means that the data conformed to the standards, formats, or rules defined for the system. For example, a phone number stored as 0123456789 is consider as valid if the system expects the value in integer format, but 011-234-56789 is invalid because it is a string.

Data Quality Dimensions	**Definitions**
Validity	i. "The degree to which the data agree with established rule" (Hassenstein & Vanella, 2022). ii. "Validity measures whether a value conforms to a preset standard" (Myers & P. Blake, 2017).

Compliance	i. "The degree to which data has attributes that adhere to standards, conventions or regulations in force and similar rules relating to data quality in a specific context of use" (Miller et al., 2024).

5. **Duplication / Uniqueness**

Duplication or uniqueness means that the data does not have redundancy in the records. For example, the key column for a dataset such as customer ID cannot appear more than once in the table.

Data Quality Dimensions	**Definitions**
Duplication	i. "A measure of unwanted duplication existing within or across systems for a particular field, record, or data set" (Sidi et al., 2012).
Uniqueness	i. "The degree to which there is no duplication in records" (Croft et al., 2023).

6. **Understandability**

Understandability means that the data is readable, easily comprehended, and expressed in appropriate languages, symbols, and units. For example, the data that has value like this '!|"W"iX P"F"W#jX"Q#' is unreadable by the user, hence the user cannot understand and make sense of the data.

Data Quality Dimensions	**Definitions**
Understandability	i. "Extent to which data are clear without ambiguity and easily comprehended" (Sidi et al., 2012). ii. "The degree to which data has attributes that enable it to be read and interpreted by users, and are expressed in appropriate languages, symbols and units in a specific context of use" (Miller et al., 2024).

7. **Amount of Data/ Quantity/ Data Volume**

This data quality dimension means that the volume of data is appropriate for the task at hand. For example, to do analytics on past year trends, the dataset must be large enough comprising all daily data collected from January to December not just aggregated data from each month.

Data Quality Dimensions	**Definitions**
Amount of Data	i. "To extent to which the quantity or volume of available data is appropriate" (Sidi et al., 2012).
Quantity	i. "The degree to which data has attributes that represent the sufficient amount or volume, providing a comprehensive view of the intended attribute of a concept or event in a specific context of use" (Miller et al., 2024).
Data Volume	i. "The number of raw data items (values) available for use to compute a result data item (in a stream query or sub-query)" (Liu et al., 2020).

8. **Confidentiality**

Confidentiality means that the data has confidential attributes which can be accessed by authorised users only. For example, the access to employee salaries data must be limited to Human Resources personnel only.

Data Quality Dimensions	**Definitions**
Confidentiality	i. "The degree to which data has attributes that ensure that it is only accessible and interpretable by authorised users in a specific context of use" (Miller et al., 2024).

9. Precision

Precision means that the data meets the exact level of details. For example, a distance value of 220.85 km is more precise than 220 km.

Data Quality Dimensions	Definitions
Precision	i. "The degree to which data has attributes that are exact or that provide discrimination in a specific context of use" (Miller et al., 2024). ii. "Precision measures the number of decimal places and rounding of a data value or level of aggregation" (Myers & P. Blake, 2017).

10. Traceability / Lineage

Traceability or lineage means that the data has attributes that provide details to track the changes made to the data and access trail to the data. For example, a log records have attributes that capture which data was updated and by whom.

Data Quality Dimensions	Definitions
Traceability	i. "The degree to which data has attributes that provide an audit trail of access to the data and of any changes made to the data in a specific context of use" (Miller et al., 2024).
Lineage	i. "Measures whether factual documentation exists about where data came from, how it was transformed, where it went and end-to-end graphical illustration" (Myers & P. Blake, 2017).

11. Portability

Portability means that the data can be migrated, moved or copy to another system while preserving the existing quality. For example, data is said to be portable if it has the exact states as in old database with no errors or missing data after it was transferred into new database.

Data Quality Dimensions	Definitions
Portability	i. "The degree to which data has attributes that enable it to be installed, replaced or moved from one system to another preserving the existing quality in a specific context of use" (Miller et al., 2024).

12. Recoverability

Recoverability means that the data can preserve or maintain to remain operational in case of failure event. For example, the data has backup or replicated in another repository.

Data Quality Dimensions	Definitions
Recoverability	i. "The degree to which data has attributes that enable it to maintain and preserve a specified level of operations and quality, even in the event of failure, in a specific context of use" (Miller et al., 2024).

13. Governance

Governance means that the data aligns with organization's rules and structures. For example, the project ID shall follow the standard naming convention as per data governance policy.

Data Quality Dimensions	Definitions
Governance	i. "The degree to which data has attributes that adhere to the formalised frameworks of authority and accountability that support harmonised data activities across an organisation" (Miller et al., 2024).

14. Semantics

Semantics means that the data has a correct, consistent, and clear meanings that is understood by users of the data. For example, the use of "DOB" as an abbreviation for Date of Birth is consistent across all systems and understood by every user.

Data Quality Dimensions	Definitions
Semantics	i. "The degree to which data accurately and consistently represents the intended meaning, interpretation, and real-world concepts within a specific context of use, ensuring correct semantic understanding by users and applications" (Miller et al., 2024).

15. Security

Security means that the data which protected under Personal Data Protection Act such as Personally Identifiable Information (PII) needs to be protected against abuse. For example, financial transactions data that contains personal data on customers such as name, email address, phone number, identity card number and bank account number must be encrypted to prevent unauthorized users from accessing it.

Data Quality Dimensions	Definitions
Security	i. "Extent to which access to information is restricted appropriately to maintain its security" (Declerck et al., 2024; Sidi et al., 2012).

	ii. “Personal data are not corrupted, and access is suitably controlled to ensure privacy and confidentiality” (Myers & P. Blake, 2017).

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